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Summary

I specialize in simplifying architectures that have grown too complex for the teams that depend on them — diagnosing where the weight is, making the structural call, and seeing it through. Over 20 years I've done this across healthcare, e-commerce, fintech, banking, and public infrastructure, in Paraguay, the US, and the Netherlands. My background starts in electrical and embedded engineering, which trained a different kind of systems instinct: I think in constraints, failure modes, and physical limits before I think in abstractions. When a system starts collapsing under growth, the root cause is almost always a physical constraint the architecture stopped respecting — spotting that gap, e.g., latency, memory pressure, is what the first half of my career trained me to see.

Experience

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UTRECHT, THE NETHERLANDS

Software Engineer

Nov '23 – Present

The core problems: a tightly-coupled legacy monolith blocking cross-team independence, an inherited culture that treated legacy stability as a reason to avoid tests, and an emerging need to govern AI-assisted development responsibly at scale.

- Migrated login handling out of the monolith; chose an event-based design over direct service coupling, enabling other teams to add post-login behaviour autonomously without coordination overhead.
- Demonstrated test-gated automatic releases on a legacy service; the approach spread organically, leading to an invitation to join the 4-person company-wide quality community.
- Owns quality and security constraints for AI-assisted code generation (Claude) company-wide; introduced promptfoo as a pre-release gate for Claude Code skills before company-wide distribution.
- Maintains systems at scale (20M+ users, 95k+ req/s) across Go, Java, Kotlin, and Python — edge router, GraphQL connectors, and the monolith-to-microfrontend migration.

Quin

AMSTERDAM, THE NETHERLANDS

Recruited as Tech Lead for the integration team; promoted to lead the platform team after identifying that the architecture itself — not the teams — was the source of the company's growth problems.

Tech Lead

Sep '21 – Oct '23

The core problems: a microservices architecture collapsing under its own weight — a feature as simple as showing a doctor their patient list took 6 to 8 months to ship, infrastructure costs exceeded \$2M annually, deployments were quarterly, and onboarding a new developer took nearly a year.

- Diagnosed the shared commons library as the most immediate drag — accumulated RestTemplate clients for every service in the landscape, loaded by every microservice regardless of need; stripped it down, cutting individual service footprint by 20%+.
- Standardized Git workflows, code review practices, and CI/CD pipelines across all backend teams.
- Bet on a modular monolith over the prevailing microservices orthodoxy; consolidated 70+ services into a single event-driven architecture with 5 bounded domains and 3 anti-corruption layers, applying Domain-Driven Design throughout.
- The consolidation cut infrastructure costs from \$2M+ to under \$300K (85%+ savings), accelerated deployments from quarterly to every merge, and reduced onboarding from nearly a year to days. The company, which had entered bankruptcy, recovered and resumed growth.

Backbase

AMSTERDAM, THE NETHERLANDS

Joined as a Backend Trainer; promoted to R&D after identifying that the product's integration approach — building scenario-specific forms for each banking workflow — would not scale.

Tech Lead

May '20 – Aug '21

The core problems: an integration strategy built around one-off scenario forms with no reusable components across them, and a release process so fragile that every deployment broke something.

- Diagnosed that the scenario forms shared underlying integration components (identity verification, SMS/email, document signing) being reimplemented repeatedly; redesigned the approach around composable connectors — personally built 5, the template grew to 12 with the team, and enabled 20+ connectors after departure.
- Diagnosed the release failures as a testing gap; introduced acceptance tests and hands-on testing practices across the team, cutting re-releases from twice weekly to once per quarter across 5 banking clients.

Backend Trainer

Jan '20 – May '20

Delivered 3 week-long training sessions on Backbase's core banking platform to approximately 30 client engineers before the role pivoted due to the pandemic.

Software Natura

ASUNCION, PARAGUAY

Consulting firm serving US clients from Paraguay. Brought in to solve a cross-system communication problem every other engineer had declined, with a one-month deadline.

Team Lead

Feb '17 – Dec '19

The core problems: a Java call center application that couldn't communicate with a COBOL mainframe (one month, declared impossible by everyone else), followed by a request to "update" the call center that warranted a full architectural diagnosis.

- Built a C-based Linux kernel module to bridge the Java/Struts application to the COBOL mainframe — the only viable path across a hard process boundary — and delivered within the one-month deadline.
- When asked to update the call center, diagnosed it as beyond patching; made the call for a full revamp in Spring Boot and Angular with domain-driven boundaries, and brought in a data specialist for Apache Spark search rather than attempting it without domain expertise.
- Diagnosed the Paraguay-side team as lacking process and standards; introduced structured workflows and code quality practices that enabled a 7-person team to collaborate effectively with 5 US-based developers across time zones.

QA Engineer

May '16 – Jan '17

Diagnosed a 100k+ line legacy CRM monolith — serving American community colleges in a 50-developer team — as having zero test coverage, meaning bugs only surfaced through weeks of end-user feedback; grew coverage from zero to 60%, shifting detection to build time.

Santco SRL

ASUNCION, PARAGUAY

Software company serving high-profile Paraguayan clients in cattle trading, logistics, and manufacturing — all with no existing software for their most critical operations and zero tolerance for downtime.

Software Engineer & Team Lead

Nov '14 – Mar '16

The core problem: clients with zero tolerance for downtime and no existing software for their most critical operations — delivery reliability, not technical capability, was the actual constraint.

- Structured a team of up to 8 developers around Agile iteration cycles, replacing ad-hoc delivery with predictable, testable releases.
- Delivered 4 process-automation applications on CakePHP and Google Cloud for Paraguay's leading cattle auction house, largest logistics company, and top leather manufacturer — the auction house trusted the software enough to increase bidding events from weekly to every two days.
- Each delivery required learning the client's domain from the ground up before writing a line of code — bidding logic, logistics routing, production cycles. That domain fluency is what made the automation trustworthy rather than just functional.

Early Career

PARAGUAY, 2005–2012

Before moving full-time into software engineering, I spent roughly seven years building physical systems at national scale: the network and data center infrastructure for Paraguay's 911-equivalent emergency services (a \$500M+ resilient optical network spanning major metropolitan areas, supporting tens of thousands of street cameras and real-time coordination of police, fire, and ambulance dispatch); a multi-tenant virtual machine provisioning platform built from first principles before AWS existed, consolidating infrastructure for 15 clients into a shared on-demand model; and network deployments across 500+ banking sites. Running alongside that: the CompuBus research project at Catholic University — embedded C and C++ on microcontrollers with Linux kernel-level communication modules, delivering a working prototype deployed with a real bus company in Asuncion. Thinking in hardware constraints before software abstractions is not background flavour — it is the lens.

Education

University of Leuven

LEUVEN, BELGIUM

Masters degree in Nanoscience and Nanotechnology

2012 – 2014

Area of interest: Bionanotechnology.

Catholic University of Asuncion

ASUNCION, PARAGUAY

Electrical Engineering (discontinued)

2007 – 2010

Bachelors in Electrical Engineering

2002 – 2006

Skills

Languages: Java, Kotlin, Go, Python, C, C++, JavaScript, PHP, SQL.

Frameworks & Libraries: Spring Boot, Netflix OSS, CakePHP, jQuery, Angular, React.

Infrastructure & DevOps: AWS, Azure, Google Cloud, Terraform, Docker, Kubernetes, Jenkins, SonarQube, GitLab CI/CD, GitHub Actions, NGINX, Apache, Tomcat, PostgreSQL, MySQL, SQLServer.

Methodologies: Agile, Scrum, Domain-Driven Design, Clean Architecture, Microservices.

Natural languages: Spanish (*native*), Guarani (*native*), English (*full professional proficiency*), German (*limited working proficiency*), French (*elementary proficiency*).

Certifications: System Thinking — MIT xPRO (2024), Nanotechnology for Health — imec (2013), Introduction to Artificial Intelligence — Udacity (2009).